Twitter Sentiment Analysis on Apple and Google Products

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Phase 4 Project

Business Problem

- Companies receive thousands of tweets about their products daily
- Manually reading and categorizing each tweet is time-consuming
- There is a need for an automated system to classify customer sentiment

Goal: Build a machine learning model that automatically classifies tweets as Positive, Negative, or Neutral sentiment

The Data

- 9,070 tweets about Apple and Google products
- Source: CrowdFlower via data.world
- 4 sentiment categories:

  - No emotion toward brand or product (5,375)

  - Positive emotion (2,970)

  - Negative emotion (569)

  - I can't tell (156)
- Dataset showed significant class imbalance

Data Preparation

- Removed 22 duplicate tweets
- Dropped 1 missing tweet
- Text cleaning:

  - Lowercased all text

  - Removed URLs, mentions (@), punctuation

  - Removed special characters
- Removed common English stopwords using NLTK
- Applied TF-IDF Vectorization (5,000 features)

Models Used

Three machine learning models were trained and evaluated:

- Model 1: Multinomial Naive Bayes

  - Simple probabilistic classifier

  - Fast and effective for text data

- Model 2: Logistic Regression

  - Linear classification model

  - Strong performance on text features

- Model 3: Random Forest Classifier

  - Ensemble of decision trees

  - Handles complex patterns

Results

- Logistic Regression achieved the best overall performance
- All models struggled with minority classes due to class imbalance

Model Comparison:

Model	Accuracy
Naive Bayes	[67.8%]
Logistic Regression	[69.3%]
Random Forest	[67.5%]

Conclusions

- Three NLP models were successfully built and evaluated
- Logistic Regression performed best with 69.3% accuracy
- All models showed bias toward the majority class due to class imbalance
- NLP-based sentiment analysis is an effective tool for automated social media monitoring at scale

Recommendations

- Use Logistic Regression as the primary sentiment classification model due to its superior performance
- Monitor customer sentiment toward Apple and Google products to identify emerging trends
- Investigate negative sentiment tweets to better understand customer concerns and areas for improvement
- Use sentiment analysis as a decision-support tool for marketing and product teams